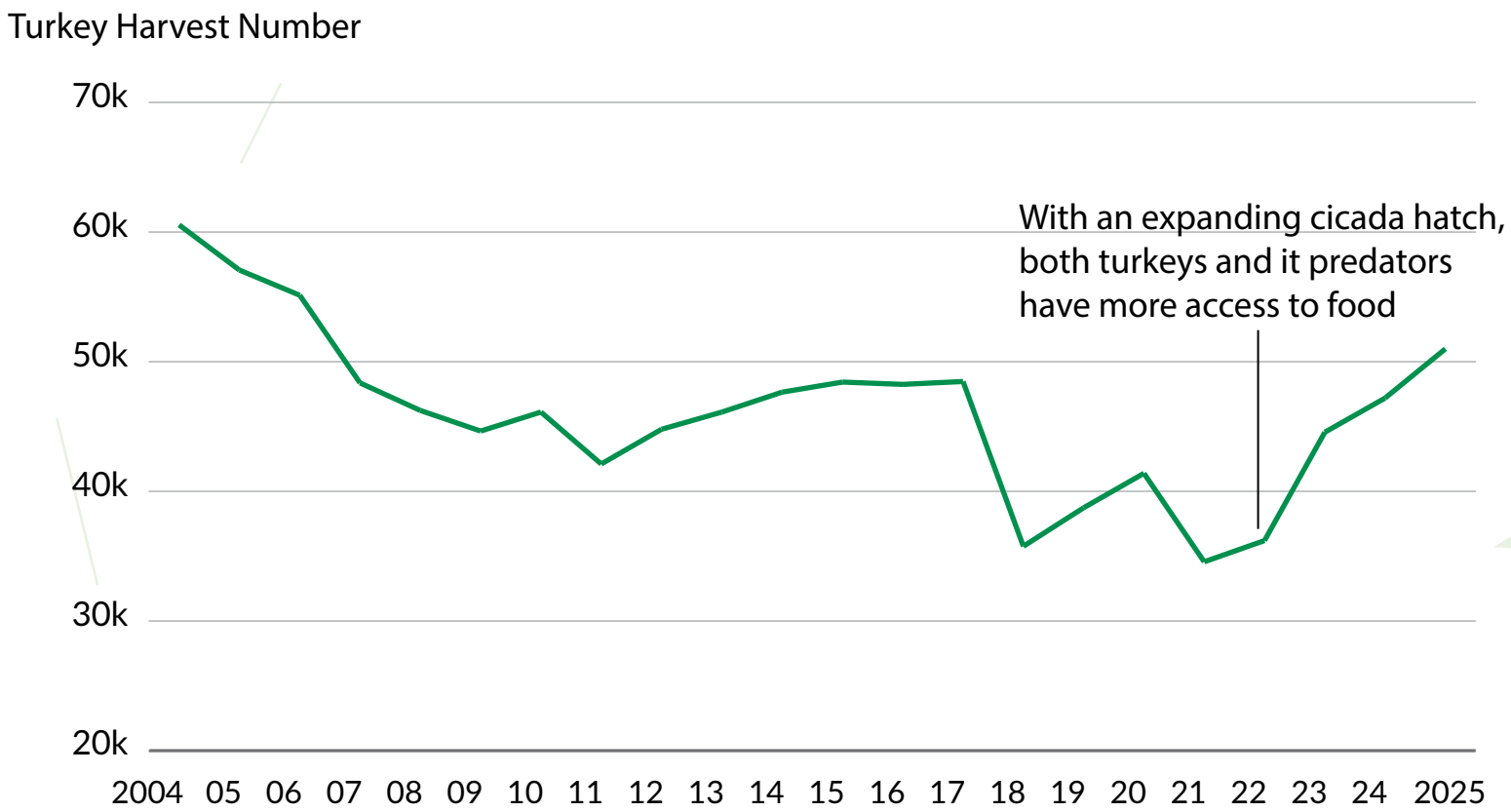
Woodland recession limits the wild turkey reproduction

By Wei He

Facing the population decline of Missouri's wild turkey in recent years, missouri Department of Conservartion(MDC)'s wild turkey program coordinator Nicholas Oakley states that the habitat change is the primary driver. Woodlands have been replaced by dense forests, which undermine the survival rate of poults (young turkeys).

Missouri's wild turkey harvest has dropped 40% from 2004 to 2025

With no formal method to count wild turkeys, Missouri relies on turkey harvest totals to gauge population health. Once nearly soared in the mid-20th century, Missouri's wild turkey population rebounded through restoration efforts and hunting management. Despite a recent increase in harvest numbers since 2023, the state still saw a **40% drop** in turkey harvest from 2004's peak to 2022.

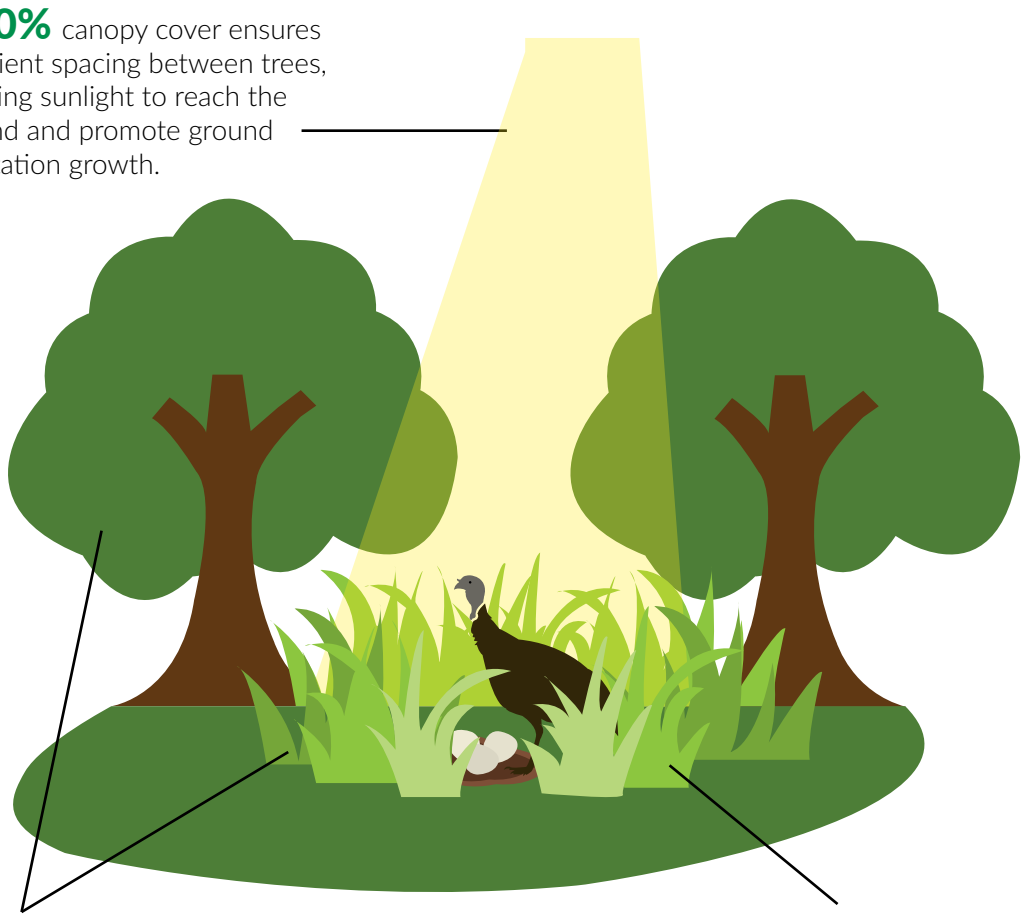


Nicholas Oakley
MDC's wild turkey program coordinator

*"There's quite a few things that impact the poult survival...Predator is not the primary driver. It is a **habitat issue**."*

An ideal nesting site

< 60% canopy cover ensures sufficient spacing between trees, allowing sunlight to reach the ground and promote ground vegetation growth.

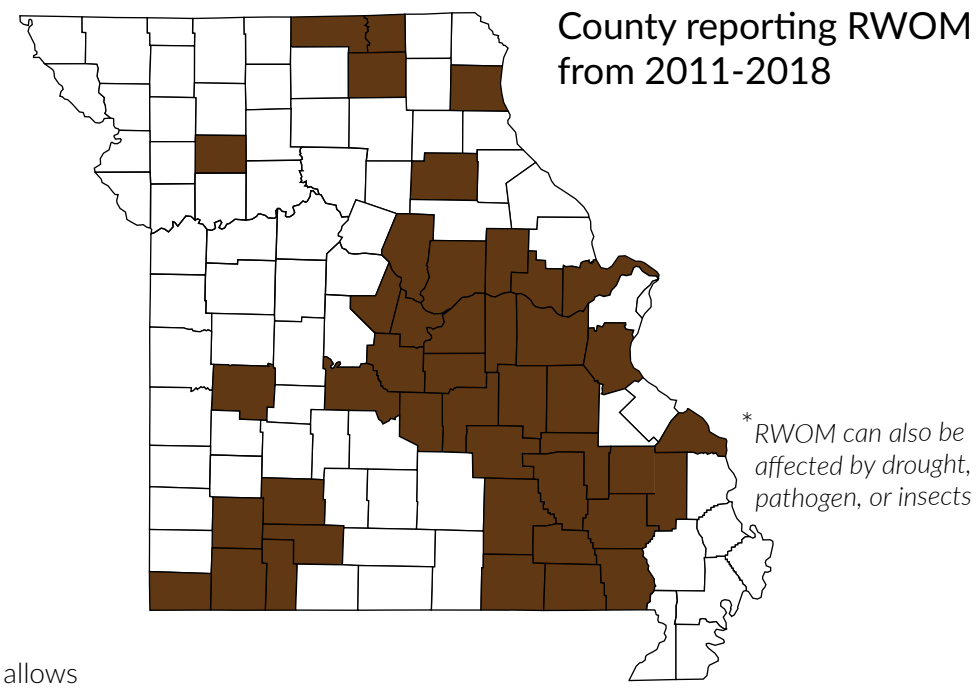


Diverse plants containing trees and grasses provide young turkeys with insects at different stages of their growth.

Waist-high vegetation that allows the hen to keep her head above the cover to watch for predators and to hide her eggs.

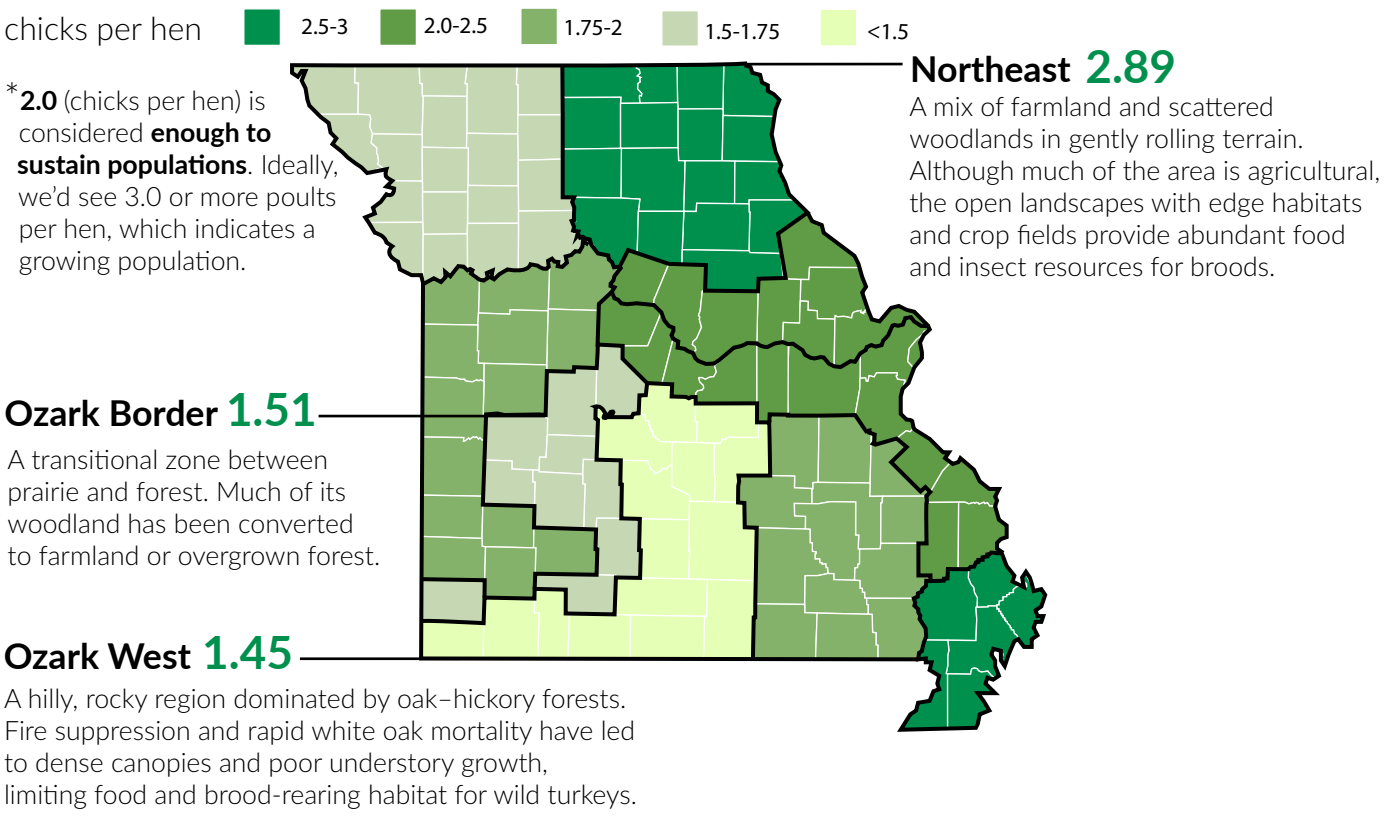
A turkey habitat change suggested by white oak recession

Rapid white-oak mortality (RWOM) across southern Missouri signals a shift from open woodland to closed-canopy forest. Historically, natural fire kept canopies below 60 %. Without it, shade-tolerant species crowd in, sunlight and ground cover vanish, and acorn crops plummet, stripping wild turkeys of nesting sites and critical fall food.



*RWOM can also be affected by drought, pathogen, or insects

Average number of chicks per hen across turkey productivity regions



How landowners create a good habitat

1 Evaluating and planning your land

- Visit the website of MDC. Fill in the Wild Turkey Habitat Evaluation Worksheet to evlauate your land.

2 Thinning the forest

- Cut mature trees can help foster a healthy habitat. The canopy gap allows ample amount of sunlight to the forest floor, thus fostering new trees and ground vegetations to grow.

3 Burning the forest

- Fire has shaped the landscape of Missouri woodland historically. It sets back the habitat to forbs and grasses. Burn the land floor to remove dense forbs and stimulate native plant growth.

***Contact a MDC staff for the help of technical advice, step-by-step recommendation, cost-share information, and burn plans.**

Source: Missouri Department of Conservation, The National Wild Turkey Federation